

OTAR, Kazakhstan

WMO: 368640

Lat: 43.533N

Lon: 75.250E

Elev: 743

StdP: 92.71

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-19.7	-26.3	0.4	-23.4	-22.7	0.5	-19.3	10.4	-2.4	8.5	-2.1	0.2	90	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	36.3	17.5	34.7	17.0	33.1	16.5	18.9	30.3	18.2	30.0	17.7	29.3	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	12.5	21.4	14.8	11.5	21.1	13.7	10.7	20.8	56.8	30.1	54.5	30.1	52.6	29.1	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.7	6.6	-26.5	39.2	4.4	1.3	-29.7	40.1	-32.3	40.9	-34.8	41.6	-38.1	42.6		
(5)			-26.7	20.8	4.4	2.0	-29.9	22.2	-32.5	23.4	-35.0	24.5	-38.2	25.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-7.1	-4.3	4.9	11.3	16.7	22.1	24.5	23.1	17.5	10.0	2.3	-5.1	(6)	
	DBStd	11.92	6.34	6.24	5.20	4.49	3.72	3.08	2.95	3.17	4.25	4.62	5.59	5.48	(7)	
	HDD10.0	1897	529	400	167	37	2	0	0	0	2	57	236	469	(8)	
	HDD18.3	3669	787	633	415	213	76	6	1	3	66	261	482	727	(9)	
	CDD10.0	1798	0	0	10	75	208	364	450	404	225	57	4	0	(10)	
	CDD18.3	528	0	0	0	2	24	120	192	149	39	2	0	0	(11)	
	CDH23.3	7583	0	0	1	56	423	1572	2659	2138	684	51	0	0	(12)	
	CDH26.7	3576	0	0	0	9	127	695	1401	1067	267	9	0	0	(13)	
(14) Wind	WSAvg	2.2	1.3	1.6	2.4	2.7	2.9	2.7	2.6	2.6	2.3	1.9	1.6	1.5	(14)	
(15) Precipitation	PrecAvg	308	18	21	26	38	37	29	27	13	14	30	31	22	(15)	
	PrecMax	492	58	62	48	85	91	69	68	41	47	89	77	42	(16)	
	PrecMin	192	6	8	10	6	5	5	6	0	0	1	4	8	(17)	
	PrecStd	76	11	11	11	20	20	17	16	12	12	23	21	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	10.1	21.6	28.1	32.5	36.5	39.2	37.7	34.4	28.0	18.5	8.0	(19)	
		MCWB	3.6	5.7	11.5	13.7	15.8	17.2	18.0	17.9	15.6	13.2	10.9	4.2	(20)	
	2%	DB	3.9	6.8	18.5	24.8	30.1	34.3	36.9	36.0	32.0	24.2	14.9	4.9	(21)	
		MCWB	1.6	3.6	10.2	13.6	15.4	16.9	17.9	17.0	14.9	12.1	9.1	2.4	(22)	
	5%	DB	2.3	4.7	15.7	22.4	27.8	32.6	35.2	34.3	29.9	21.5	12.2	3.3	(23)	
		MCWB	0.6	2.3	9.2	12.8	15.0	16.5	17.4	16.6	14.2	10.9	7.5	1.2	(24)	
	10%	DB	0.9	3.0	13.2	20.0	25.7	30.7	33.4	32.3	27.5	18.6	9.9	1.8	(25)	
		MCWB	-0.4	1.1	8.1	11.8	14.3	16.2	17.0	16.2	13.5	9.8	6.7	0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	6.2	12.7	16.5	18.4	19.4	20.1	19.2	16.9	14.2	11.8	4.8	(27)	
		MCDB	6.8	9.5	18.8	24.0	27.2	29.4	30.5	31.0	28.7	24.5	17.0	6.9	(28)	
	2%	WB	2.1	4.0	11.0	14.8	17.0	18.5	19.1	18.2	15.8	13.0	9.9	2.7	(29)	
		MCDB	3.6	6.5	17.1	21.9	25.9	28.4	31.0	31.4	28.8	21.4	13.7	4.5	(30)	
	5%	WB	0.8	2.4	9.6	13.8	16.0	17.8	18.4	17.6	14.9	12.0	8.2	1.4	(31)	
		MCDB	2.1	4.4	14.8	20.8	24.6	27.8	30.8	30.7	27.4	19.3	11.4	3.0	(32)	
	10%	WB	-0.3	1.2	8.4	12.6	15.2	17.2	17.8	16.9	14.1	10.8	6.6	0.3	(33)	
		MCDB	0.9	2.8	13.0	18.8	23.6	27.3	30.2	29.7	26.1	17.1	9.7	1.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	8.5	10.5	12.0	13.9	14.5	15.1	15.8	15.8	12.8	9.3	8.5	(35)	
		MCDBR	10.0	10.1	15.0	16.9	17.5	17.7	18.4	18.8	19.3	17.9	14.0	10.5	(36)	
	5% WB	MCWBR	7.8	7.2	8.2	7.8	6.5	5.4	5.8	6.4	7.5	8.0	8.4	8.1	(37)	
		MCDWR	8.6	9.2	13.5	14.8	14.5	14.7	15.2	16.3	16.7	14.3	12.3	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.363	0.413	0.473	0.502	0.475	0.472	0.464	0.429	0.411	0.439	0.410	0.357	(40)	
		taud	2.039	1.953	1.873	1.850	1.976	2.029	2.067	2.147	2.132	1.970	1.983	2.048	(41)	
	Ebn at Noon	Ebn at Noon	715	742	749	766	805	807	808	825	806	703	647	676	(42)	
		Edh at Noon	110	143	176	195	177	168	160	143	134	139	116	99	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.98	2.87	3.81	5.09	6.53	6.91	6.88	6.38	5.07	3.23	1.93	1.52	(44)	
		RadStd	0.21	0.32	0.32	0.46	0.45	0.33	0.41	0.31	0.22	0.30	0.19	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.48	-0.91	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page